

Ching-Han, Cheng

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OBJECTIVE

Seeking a challenging position in R&D Substitute Service to apply my expertise in Linux kernel and network programming, built through hands-on projects in OS internals, kernel modules, and low-level debugging tools, to develop reliable, high-performance system software.

EDUCATION

- **National Yang Ming Chiao Tung University (NYCU)** Feb. 2024 - Jul. 2026
Institute of Network Engineering
◦ Advisor: [Li-Hsing Yen](#)
◦ Thesis Title: Decentralized Workload Offloading for Carbon Reduction and Queue Stability in a Hybrid Cloud-Edge Environment [🔗]
 - * Developed an edge-cloud offloading framework to dynamically configure transmission power and workloads
 - * Minimized the system's carbon footprint while optimizing load balancing across edge and cloud servers
- GPA: 4.15/4.30
- **National Sun Yat-Sen University (NSYSU)** Sep. 2020 - Jan. 2024
Department of Computer Science and Engineering
◦ Ranking: 5/66
◦ GPA: 4.08/4.30 Kaohsiung, Taiwan

PROJECTS

- **xv6-riscv Kernel: [MIT 6.1810 Operating Systems Labs] (Self-Study, In Progress)** Jul. 2026 - Present
Tools: [C, RISC-V, GDB, QEMU] [🔗]
 - Independently completed MIT 6.1810's first 5 kernel labs, then resolved cross-subsystem merge conflicts (page tables, copy-on-write) to unify them into one regression-free kernel
 - Designed a two-tier (4KB page / 2MB superpage) buddy-style physical memory allocator with automatic split/coalesce, and extended the *sbrk* syscall with eager/lazy and normal/super allocation flags
- **Single Instruction Debugger: [NYCU Advanced Programming in the UNIX Environment]** spring 2025
Tools: [C++, Capstone, ptrace] [🔗]
 - Built an instruction-level debugger, validated by correct breakpoint handling and syscall tracing
 - Implemented *ptrace*-based process control, memory patching, and disassembly support
- **Remote Working Ground Server: [NYCU Network Programming]** spring 2025
Tools: [C++] [🔗] Project 2, 3]
 - Developed a multi-client shell server, and extended built-in commands (broadcast message and firewall)
 - Project 2: Implemented single-process concurrent connections and user-to-user pipe communication
 - Project 3: Implemented concurrent connection-oriented paradigm, per-user environment management and shared memory synchronization by integrating *signals*, and *FIFO*-based user pipes
- **AES Linux Kernel Module: [NYCU Advanced Programming in the UNIX Environment]** spring 2025
Tools: [C, QEMU, Docker] [🔗]
 - Implemented a Linux kernel module with AES encryption/decryption functionality
 - Supported read, write, and *ioctl* system calls, and verified concurrent read-write operation and thread safety

TEACHING ASSISTANT EXPERIENCE

- **Introduction to Network Programming** fall 2025
NYCU
 - Assisted the instructor in reviewing and refining lab procedures
 - Refactored and customized shell-based automated testing scripts using *tmux* to accelerate student assignment testing and streamline grading workflows [📺 Together] [📺 Separately]

HONORS AND AWARDS

- **EXCELLENT STUDENT AWARDS (× 2)** fall 2020, spring 2021
NSYSU
 - Top 5% students
- **Excellent Performance in Professional Courses (× 3)** fall 2021, spring 2022, fall 2022
NSYSU
 - Received A+ grades in three core professional courses

CERTIFICATIONS

- **TOEIC: 805 points** Aug. 2025
- **Collegiate Programming Examination (CPE): Top 0.2% (4th/1,814), solving 6 out of 7 problems** May 2022